A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Habitat case studies: woodland, rainforest, reefs, islands, mangroves, Antarctica - Set 1

Mark scheme

AQA spec 3.1.2.3.4; The living environment; textbook pp. 50-71

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** No major temperature extremes – regular water supply with no pronounced dry season – deep fertile soils built by leaf litter and root binding – strong seasonality **1** Deciduous canopy present spring to autumn shades the floor during the best growing temperatures **1** Bluebells, snowdrops, lesser celandine grow and flower early in spring before the trees leaf **1** Winter food shortage causes animals to hibernate (dormice), store food or migrate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Long time for species to colonise – some woodland plants colonise very slowly so their presence (ancient woodland indicators) shows the wood is old **1** Natural woodland has trees of all ages with clearings where old trees died, giving varied light and dead wood habitats **1** A new plantation has a single-age dense canopy with low light and few niches until trees start to die (over 200 years) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	3
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Clearance for farmland, plantations, urban development, roads, mineral extraction **1** Fragmentation isolating remaining woods so populations cannot move between them and are vulnerable **1 Management change:** end of coppicing so woods become uniform and dark – replacement by conifer monocultures **1** Overgrazing by deer preventing regeneration – introduced pathogens (ash dieback) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Marks shared** Up to 5 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2** A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds – oak supports many invertebrates **up to 2** B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies – fewer niches **up to 2** C: non-indigenous single-species, single-age conifers support few native invertebrates – dense shade – no clearings or dead wood – needles acidify soil **up to 2** Habitat diversity / structural variety increases the range of niches and hence species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	5
02.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Designated protected areas:** SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks **1 Legal protection of ancient woodland:** planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) **1 Conservation management:** coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer **1** Planting new native woodlands linking fragments (corridors) – Woodland Trust / National Trust ownership **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** No species is completely dominant and food webs have many alternative links **1** So a change in the population of one species has a relatively small impact on the whole community **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors – many rare species with small ranges **1** High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) **1** Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches – intense competition for light drives specialisation **1** Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships – plants flower and fruit at different times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	4
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Adapted to stable conditions so they have narrow ranges of tolerance – a small change can move conditions outside the range **1** Many species are rare with small ranges / small populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare) **1** Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Over half the Amazon's rainfall comes from transpiration within the forest – clearance reduces downwind rainfall **1** Rapid decay recycles nutrients so soils are shallow / poor – canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Small clearings were farmed for a few years then abandoned as nutrients declined – the forest recolonised and soil fertility recovered before the area was cleared again **1** Population growth shortens the gap between clearances so the forest and soil do not recover **1** Clearings enlarge and degrade – less regrowth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Rainforest soils are thin and nutrient-poor (nutrients held in the biomass) – once cleared, fertility falls in a few years and yields drop, so more land is cleared **1** Abandoned land may not be recolonised if no forest survives nearby – erosion of exposed soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing **1** Reservoirs (eg Tucurui, 2850 km^2) flood large areas, block migratory fish and river dolphins – decomposing vegetation produces acidic, deoxygenated water – fluctuating downstream flows **1** Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees) – cloud forests lose humidity as clouds form higher so frogs and epiphytes decline – some models predict replacement by savannah **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt **1** In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Protected areas:** Korup NP Cameroon (1260 km^2, 600 tree species, endemic Ancistrocladus korupensis with anti-HIV / antimalarial alkaloids) – Tumucumaque NP Brazil (39 000 km^2, adjoining French Guiana reserve) – Alto Maues – Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps – sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation – ITTO aims (largely unmet) – CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia) – community projects and employment – carbon sequestration payments • **Effectiveness:** protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict – drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest – climate change cannot be controlled locally • **Judgement:** combination of large protected areas, economic alternatives for local people and reduced international demand needed – overall loss continues **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **endemic** → found naturally only in that one place • **sustainable** → meeting present needs without stopping future generations meeting theirs • **CITES** → the treaty that controls international trade in endangered species	9
	Do not credit • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Protected areas are the core of rainforest conservation. Korup National Park in Cameroon protects 1260 km^2 with about 600 tree species, including the endemic *Ancistrocladus korupensis* whose alkaloids have anti-HIV and antimalarial activity; Brazil's Tumucumaque National Park covers 39 000 km^2 and adjoins a reserve in French Guiana; the Rio Bravo area in Belize is funded by charities, companies, ecotourism and selective logging of high-value mahogany that does not damage the forest long term. Debt-for-nature swaps have cancelled Guatemalan and Brazilian debt in return for protection, CITES listing restricts trade in Honduras mahogany, the ITTO sets sustainable timber aims, ecotourism and community employment give local people an income from standing forest, and carbon sequestration payments are emerging.	
	Where they are enforced, protected areas work, but they face illegal logging, mining, poaching, road building and conflict, and ecotourism itself has damaged sites in Costa Rica and Malaysia. The deeper problem is that the drivers of deforestation - population growth and international demand for beef, soya, palm oil, timber and minerals - originate outside the forest, and climate change threatens even well-protected areas; the ITTO's aims are largely unmet and overall loss continues.	
	The most effective schemes are those that combine large protected areas with economic alternatives for local people, such as Rio Bravo, but they can only slow the loss unless demand in importing countries also falls.	
	Weaker answer - Level 2 (4 marks)	
	Rainforests are protected by national parks where logging is banned, for example in Brazil. Debt-for-nature swaps mean a country's debt is cancelled if it protects its forest. Ecotourism brings money from tourists so the forest is worth more standing than cut down. Sustainable logging only takes some trees. These efforts have saved some areas but deforestation is still happening because of illegal logging and farming, so they have only been partly effective.	
	What makes the difference	
	• Level 2 covers the main types of effort and a fair conclusion, but the question says 'describe' - Level 3 gives named examples with figures (Korup, Tumucumaque, Rio Bravo) as the mark scheme expects.	
	• Level 3 explains *why* effectiveness is limited (drivers originate outside the forest, ITTO aims unmet, climate change) rather than just noting continued loss.	
	• The judgement in Level 3 identifies what works best and what would be needed; Level 2 stops at 'partly effective'.	
05.2	1 mark for the point below. **1** Carbon sequestration reducing climate change – genetic resources / medicines – rainfall regulation – global biodiversity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm^2) **1** They use the polyp's CO2 and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently **1** They supply the polyp with glucose and oxygen - over 95% of its food energy – the polyp provides a protected habitat and nutrients **1** Polyps also trap plankton with cilia and stinging nematocysts – surplus CO2 is used to build the calcium carbonate skeleton **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
06.2	**1 mark each** 1 mark for each of the 3 points below. **1** High light (shallow, clear water) for the algae – warm stable temperatures (about 18-30 °C) – low turbidity – constant salinity – well-oxygenated water **1** Found only where all these coincide (about 1% of ocean area) **1** Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **range of tolerance** → the range of conditions in which a species can survive	3
06.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Fisheries:** support over a quarter of marine species including fish, lobsters, conch **1 Erosion protection:** complex structure dissipates wave energy protecting low coral islands – coral sand builds shores **1 Medicinal discoveries:** AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts **1 Climate control:** carbonate skeletons sequester CO2 **1** Tourism (diving, snorkelling) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Overall decline from 62% to 22% **1** Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**1 mark each** 1 mark for each of the 4 points below. **1** The polyps expel their symbiotic zooxanthellae so the coral loses colour – bleached corals are alive but grow very slowly and may die if algae do not return **1** Raised water temperature (climate change / El Niño) **1 Pollution:** sewage, pesticides, cyanide, sunscreen – low oxygen from zooplankton blooms after overfishing **1** High light / solar activity – ocean acidification hinders skeleton formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps **1** Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release **1** Physical damage from anchors, divers, souvenir collection – loss of mangroves / seagrass nurseries that trap sediment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
07.4	1 mark for the point below. **1** Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps **1** Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
08.2	**1 mark each** 1 mark for each of the 4 points below. **1** Fixed mooring buoys so boats do not anchor on coral **1** Education of divers / snorkellers not to touch coral or kick up sediment – bans on spear fishing and souvenir collection **1 Fishing controls:** no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling **1** Protection of turtle nesting beaches – control of coastal development – protection of mangroves and seagrass – artificial reefs (reef balls, sunk ships / oil rigs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** No sunlight so no symbiotic algae – polyps rely only on captured food so have far less energy and grow very slowly **1** Damage from trawls takes hundreds of years to repair – the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit	2

Do not write outside the box

Question	Marking guidance	Marks
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Deep-water trawling (also by-catch) – oil and gas exploration in deeper water – ocean acidification **1** Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	

Do not write outside the box